CLAIMS

What is claimed is:

1. A computer data bus apparatus for transmitting data words between components, comprising:

- a plurality of parallel optical channels; and
- a controller configured to cause a single data word with a bit-width greater than 128 bits to be transmitted in a single cycle by encoding portions of the data word across the plurality of parallel optical channels.

2. The apparatus of claim 1, wherein the bit-width of the data word is 256 bits.

3. The apparatus of claim 1, wherein the bit-width of the data word is 1024 bits.

4. The apparatus of claim 1, wherein each parallel optical channel is a pixel within a two-dimensional array of pixels.

5. The apparatus of claim 4, wherein a 256-bit data word is transmitted using 11 or fewer pixels.

6. The apparatus of claim 4, wherein a 1024-bit data word is transmitted using 43 or fewer pixels.

7. The apparatus of claim 1, wherein the bus is an implementation of the optical interface described in U.S. Patent Application No. 19/559,705.